Detecting Backdoor-Triggered Inferences from Hardware Performance Counters with a One-Class Anomaly Ensemble

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***Abstract*—A backdoored neural network behaves normally on clean inputs but misbehaves when a trigger pattern is present. We address detecting triggered inferences from hardware performance counter (HPC) traces when only 800 benign traces (cache-references, cycles, LLC-loads) are available and no backdoor trace is ever seen. We train a one-class ensemble of four detectors (robust Mahalanobis distance, Isolation Forest, One-Class SVM, Local Outlier Factor) whose rank-normalized scores are averaged, and calibrate the decision threshold out-of-sample by 5-fold cross-validation on benign data to a 1% target false-positive rate. Because real backdoor traces are unavailable, detection power is probed only with synthetic perturbations of benign traces; these show that the ensemble flags uniform upward shifts of 5% or more at least 99% of the time but a 5% uniform decrease only 32%, flags a weak single-counter spike 52% of the time (versus 5% for a per-feature z-score rule), is nearly blind to sub-1% shifts, and over-reacts to pure noise. True positive rate on real backdoor traces is unverified.**

***Index Terms*—backdoor attacks, hardware performance counters, anomaly detection, one-class learning, ensemble**

I. Introduction

In a backdoor attack, an untrusted party trains a model so that it behaves normally on most inputs but changes its prediction when a special pattern is embedded in the input [1]. Processing a triggered input may perturb microarchitectural activity, which HPCs expose cheaply at run time; HPC traces have previously been used to detect malware [2]. This work studies a strict anomaly-detection setting: the detector is trained on HPC traces from benign inferences only, and must label traces from backdoor inferences differently at test time.

The setting rules out supervised training and, crucially, rules out measuring detection power on real attacks. Our contributions are therefore (i) a one-class ensemble detector with an out-of-sample calibrated threshold, and (ii) a deliberately adversarial self-evaluation on synthetic perturbations that documents where the detector is weak, rather than a single favorable accuracy number.

II. Data

The training set is a CSV of 800 benign inferences with three counters: `cache-references`, `cycles`, `LLC-loads`. There are no missing values. `cache-references` and `LLC-loads` are strongly correlated (r=0.92); `cycles` correlates weakly with both ($r \approx 0.53$). `cycles` is right-skewed with heavy tails (skew 0.90, excess kurtosis 6.76, maximum $|z|$=8.5), and three rows exceed $|z|$=3 in some counter. This motivates a robust covariance estimator and an ensemble rather than a single Gaussian model. Counters are standardized to zero mean and unit variance.

III. Method

A. Base detectors

Four one-class models are fit on the standardized benign data, each producing a score where larger means more anomalous: (1) robust Mahalanobis distance from a Minimum Covariance Determinant fit [3] (support fraction 0.9), which is not dragged by the heavy tail of `cycles`; (2) Isolation Forest [4] (300 trees); (3) One-Class SVM [5] (RBF kernel, ν=0.05); and (4) Local Outlier Factor [6] in novelty mode (k=20). The first three reason about deviation from one global model of normal; LOF instead compares a point's density to its own neighborhood. All models use scikit-learn [7].

B. Score fusion

Scores from different models are not comparable, so each is mapped to its percentile rank among that model's training scores S_k:

$$r_k(x) = \frac{|\{s \in S_k : s \le s_k(x)\}|}{|S_k|}, \qquad s(x) = \tfrac{1}{4}\sum_{k=1}^{4} r_k(x). \tag{1}$$

A trace is flagged as backdoor when $s(x) \ge \tau$. Averaging hedges against any single model's assumption being wrong, which cannot be checked without attack data.

C. Out-of-sample threshold calibration

Choosing τ from in-sample scores would understate the false-positive rate on unseen benign traces. Instead, for each of five folds the ensemble is fit on four folds and scores the held-out fold; the held-out scores are pooled and τ is set to their $(1-\alpha)$ quantile, with target α=1%. The final models are then refit on all 800 traces and use this τ. The cross-validated FPR at τ=0.9778 is 1.00%, and Fig. 2 shows the benign score distribution.

IV. Evaluation

A. Benign generalization

As an independent check, the pipeline was retrained on a random 85% of the traces (680) and applied to the remaining 120 unseen benign traces; 2 were flagged (1.67%), consistent with the 1% target given the small held-out sample.

B. Synthetic sensitivity probes

No backdoor trace exists, so detection power is probed by perturbing 100 benign traces in different ways and measuring the flagged fraction. These perturbations are constructed by the authors and are *not* real triggers; they show how the detector reacts, not what its true positive rate is. Table I uses a uniform scale-up of all counters (LLC-loads by 1.5× and cycles by 0.5× the nominal percentage), and Fig. 3 sweeps other shapes.

TABLE I
Fraction flagged for a uniform proportional shift (n=100)

Shift	0% (benign)	2%	5%	10%	20%
Flagged (%)	1.0	20	99	100	100

Three findings stand out. First, sub-1% shifts are nearly invisible (2–4% flagged), so an adversary tuning a trigger to leave a tiny

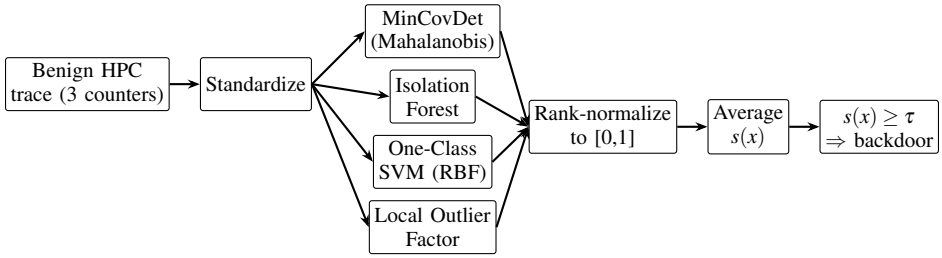


Fig. 1. Detection pipeline. Four one-class models are fit on benign data only; their scores are rank-normalized and averaged, and the threshold τ is calibrated out-of-sample.

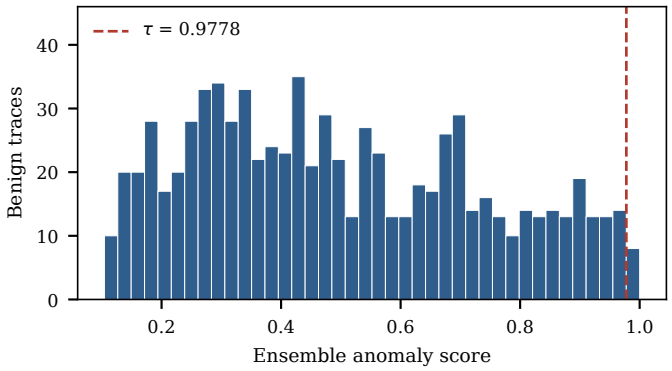


Fig. 2. Ensemble score over the 800 benign training traces and the calibrated threshold.

TABLE II
ENSEMBLE VS. PER-FEATURE Z-SCORE BASELINE, FLAGGED (%)

Perturbation	z-score	Ensemble
Uniform +10%	100	100
LLC-only +5%	5	52
Stealthy +0.5%	2	2

counter footprint could plausibly evade detection at this operating point. Second, a 5% LLC-only spike, which breaks the usual cross-counter correlation, is flagged 52% of the time, while a 5% uniform decrease is flagged only 32% (versus 99% for a 5% increase), so direction matters. Third, and unfavorably, pure noise with no attack signal is flagged 32%, 79% and 98% of the time at 1%, 3% and 6% jitter: the detector responds to deviation magnitude, not specifically to attack-like structure, so a noisier collection environment could inflate false positives well beyond the calibrated 1%.

C. Baseline comparison

To test whether the ensemble's complexity is warranted, it was compared against a per-feature z-score rule (flag if any counter's $|z|$ exceeds a threshold), calibrated to the same 1% FPR by the same cross-validation (Table II). Both methods saturate on a strong shift and both are blind to a 0.5% shift; the ensemble's advantage is confined to the weak single-counter case, where it detects about ten times as often.

D. Feature-set ablation

Three scale-invariant ratio features (cycles/cache-references, LLC-loads/cache-references, LLC-loads/cycles) were tested on the hypothesis that a trigger alters the relation between counters more than their magnitude. With the full four-model ensemble the hypothesis failed

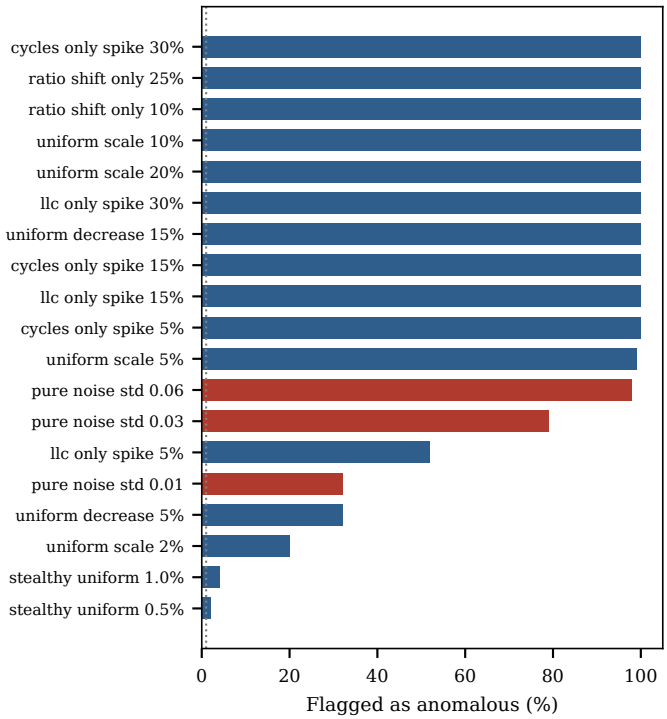


Fig. 3. Flagged fraction across synthetic perturbation shapes (n=100 each). Red bars are pure random jitter with no directional attack signal. Dotted line: 1% nominal FPR.

TABLE III
FEATURE-SET ABLATION (4-MODEL ENSEMBLE), FLAGGED (%)

Features	Uniform +10%	LLC +5%	Ratio shift
Raw (3), shipped	100	52	100
Raw + ratios (6)	100	25	100

(Table III): raw counters alone detect the weak LLC-only spike more than twice as often (52% vs. 25%) and tie elsewhere. A plausible but untested explanation is that three extra, partly collinear dimensions dilute LOF's local-density estimate. The shipped model therefore uses the three raw counters only.

E. Operating-point sensitivity

A grid over contamination $\in \{0.01, 0.03, 0.05, 0.10\}$, $\nu \in \{0.01, 0.03, 0.05, 0.10\}$ and $\alpha \in \{1, 2, 5\}\%$ shows that contamination and ν barely matter, because rank normalization removes their influence on the final score scale; α is the real lever (Fig. 4). Relaxing α from 1% to 5% raises weak-signal detection from 51.5% to

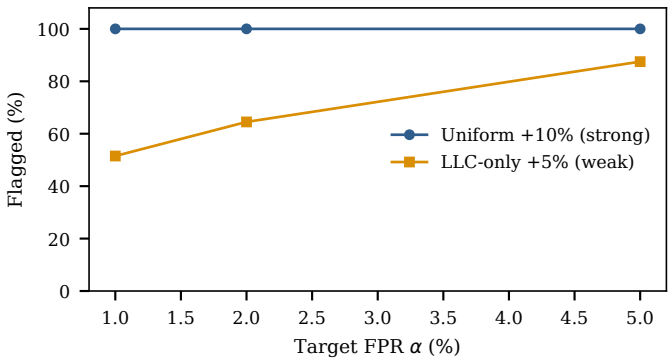


Fig. 4. Detection of a strong and a weak synthetic signal as the target FPR is relaxed.

87.5% (means over the 16 contamination/ν settings) at five times the benign false-positive rate. We keep α=1% because Accuracy, F1 and AUROC all penalize false positives; α is a command-line option if the evaluation mix differs.

V. Limitations

- **True positive rate is unverified.** Every detection figure above comes from synthetic perturbations of benign data that we constructed. They show direction of response, not performance on real triggers.
- **Stealthy triggers may evade detection** (sub-1% shifts flagged 2–4%).
- **Specificity–sensitivity trade-off.** Adding LOF and dropping ratio features raised weak-signal detection but also raised reaction to benign noise relative to an earlier raw+ratio three-model version, so real false-positive rate may exceed 1% in a noisier environment.
- **Rank normalization is relative to the training distribution.** If the evaluation traces come from a different CPU or scheduling environment, clean traces may sit differently against the reference scores; the cross-validated threshold only partly mitigates this. Batch-level normalization or an absolute threshold would be more robust.
- **Only three counters and one collection run** were available; richer counters (branch and TLB events) would likely improve separability.

VI. Conclusion

A four-model one-class ensemble with a cross-validated threshold gives a well-calibrated benign false-positive rate (1.00% cross-validated, 1.67% on 120 unseen traces) from only 800 benign HPC traces. Synthetic probes show it reacts strongly to upward counter shifts and, unlike a per-feature baseline, partly detects single-counter anomalies, but they also expose evasion by tiny shifts and sensitivity to noise. Its performance on real backdoor traces remains to be measured on labeled data; the repository includes an evaluation script for Accuracy, TPR, FPR, F1 and AUROC once such data is available. Code, the saved model and all scripts reproducing these results are in the accompanying repository.

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